# Python Activity-Based Learning<br/>Case Study

# 7 Retail Billing with Multiple Discount Rules Selected case study | Prepared using the topics listed in the supplied assignment. 1. Problem Statement

Prepare an itemized retail bill for several products. Calculate quantity-based totals, apply a threshold discount, add tax, and print the final invoice.

## 2. Objectives

• Store product details in lists.

• Use loops to accept multiple items.

• Calculate subtotals, discounts, tax, and final amount.

• Apply conditional discount rules.

## 3. Assumptions

• The customer can purchase multiple products.

• Each product has a name, unit price, and quantity.

• A 10% discount is applied when the subtotal is at least ■2,000.

• Tax is 5% of the amount after discount.

• Prices and quantities must be positive.

## 4. Algorithm

1. Initialize an empty item list and subtotal.

2. Read the number of products.

3. For each product, read name, price, and quantity.

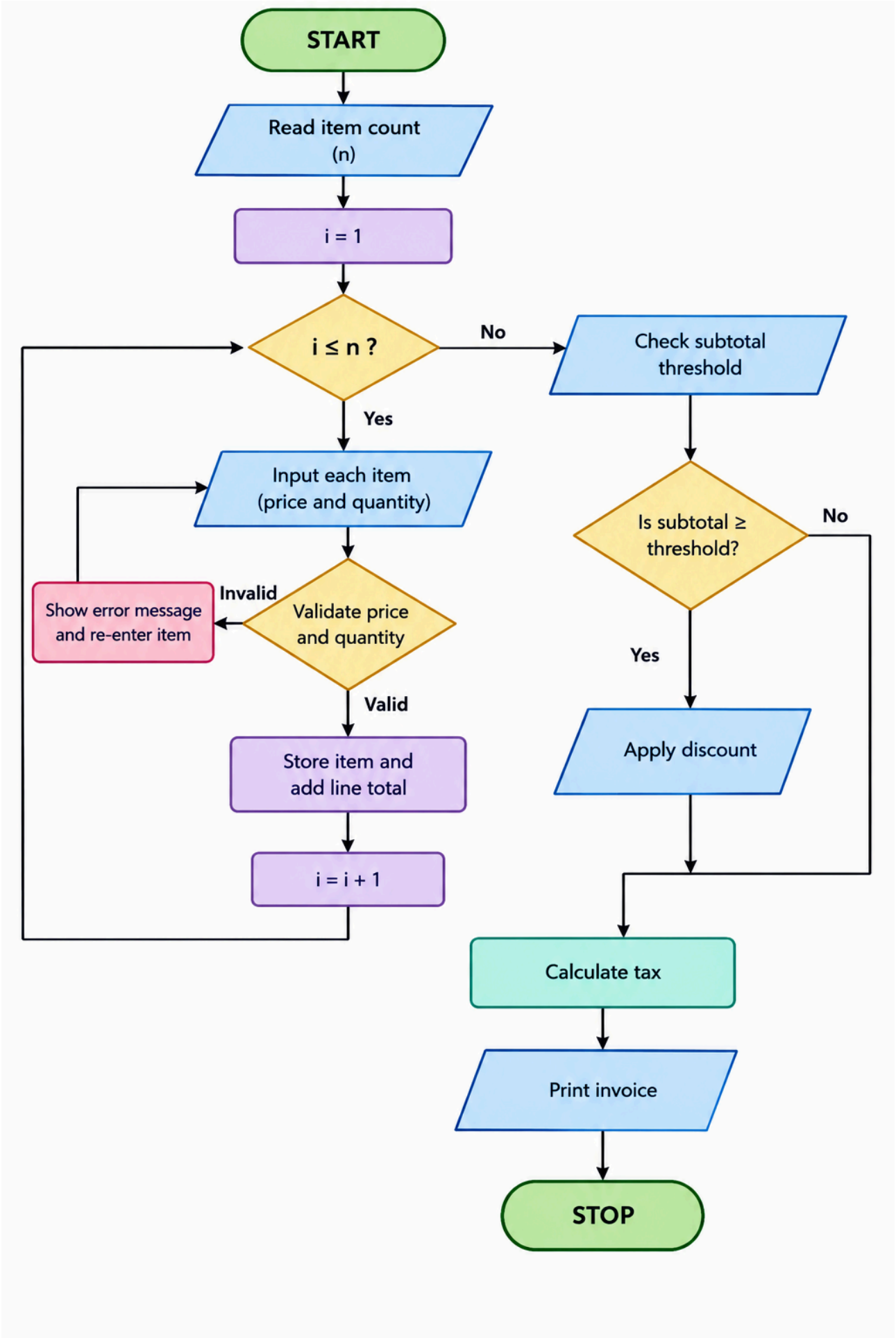
4. Validate price and quantity; accept valid entries and calculate line total.

5. Calculate discount based on the subtotal threshold.

6. Calculate tax on the discounted amount.

7. Print item details, subtotal, discount, tax, and final total.

## 5. Flowchart


START
Read item count (n)
i = 1
i ≤ n ?
No
Yes
Input each item (price and quantity)
Validate price and quantity
Invalid
Show error message and re-enter item
Valid
Store item and add line total
i = i + 1
Check subtotal threshold
Is subtotal ≥ threshold?
No
Yes
Apply discount
Calculate tax
Print invoice
STOP

## 6. Pseudocode

```
BEGIN
 subtotal ← 0
 FOR each item
 INPUT name, price, quantity
 IF price > 0 AND quantity > 0 THEN
 line_total ← price × quantity
 subtotal ← subtotal + line_total
 STORE item details
 END IF
 END FOR
 IF subtotal >= 2000 THEN discount ← 10% of subtotal
 ELSE discount ← 0
 taxable ← subtotal - discount
 tax ← 5% of taxable
 PRINT invoice and taxable + tax
END
```

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## 7. Python Source Code

```
items = []
subtotal = 0

n = int(input("Number of products: "))

for i in range(n):
 name = input("Product name: ")
 price = float(input("Unit price: "))
 quantity = int(input("Quantity: "))

 if price > 0 and quantity > 0:
 line_total = price * quantity
 items.append([name, price, quantity, line_total])
 subtotal += line_total
 else:
 print("Invalid item. It is not added.")

if subtotal >= 2000:
 discount = subtotal * 0.10
else:
 discount = 0

taxable_amount = subtotal - discount
tax = taxable_amount * 0.05
grand_total = taxable_amount + tax

print("\n--- RETAIL INVOICE ---")
for item in items:
 print(item[0], "Price:", item[1],
 "Qty:", item[2], "Total:", item[3])

print("Subtotal:", subtotal)
print("Discount:", discount)
print("Tax (5%):", tax)
print("Grand total:", grand_total)
```

## 8. Test Cases

| Input / Scenario | Expected result |
|---|---|
| Notebook ■100 × 5; Bag ■500 × 3 | Subtotal ■2,000; discount ■200; tax ■90; total ■1,890 |
| Pen ■20 × 5 | Subtotal ■100; discount ■0; tax ■5; total ■105 |
| Item price 0 | Item rejected; not included in bill |

## 9. Conclusion

The program applies list operations, loops, arithmetic, and conditional discount rules to generate a retail invoice. Note: Run the program in Python 3. The sample test cases are illustrative; enter the values when prompted.